ORIGINAL PAPER

Morphometric Study of Lumbar Spinal Canal Dimensions in Bosnia and Herzegovina Population

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doi: 10.5455/aim.2023.31.200-205
ACTA INFORM MED. 2023, 31(3): 200-205
Received: JUL 15, 2023
Accepted: SEP 07, 2023

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ABSTRACT

Background: The lumbar spinal canal consists of 5 interconnected lumbar vertebrae through which the final part of the spinal cord passes and the lumbar and sacral spinal nerves that form the cauda equina. The lumbar canal stenosis can directly affect neurological symptoms and pain in the lumbar region and lower extremities. Due to the frequency of such symptoms, lumbar stenosis has been the subject of research around the world. **Objective**: The objective of this study was to measure, analyze and compare the mediosagital and interpeduncular diameters of the lumbar spinal canal in the population of Bosnia and Herzegovina to other populations around the world. **Methods:** We conducted a retrospective descriptive study on patients (n=200) who underwent Multi-slice computer tomography (MSCT) performed on a 40-slice CT scanner (Siemens Somatom Definition AS) for lumbar pain between January 1, 2013 and December 31, 2014. Age, gender, midsagittal (MSG) and interpeduncular (IP) diameters of the lumbar spine were recorded for each patient. **Results:** Results of our study show that the largest MSG diameter is at L1 level for both sexes, with an average length of 19,06mm, and the smallest at L3 level, with an average length of 16,66mm. Our study shows that the MSG diameter is significantly larger in females than males on all levels from L1 to L5. In both sexes, MSG diameter shows the form of an hourglass with narrowing at L3 level. IPD is largest at L5 level for both sexes, with an average length of 31,94mm, and the smallest average length at L1 level, at 24,78mm. IPD diameter is significantly larger in males than females on all levels from L1 to L5. IPD shows a tendency of growth from L1 to L5 in both sexes. **Conclusion:** There were significant differences in the dimensions of the lumbar spine canal between female and male patients. We found significant difference in MSG and IP diameters measurements between Bosnian and Herzegovinian population compared to other populations. The dimensions of the lumbar canal provide a baseline of normative data for the evaluation of patients presenting with lower back pain and lumbar canal stenosis in study population.

Keywords: psychometric properties, reliability, validity, responsiveness, outcome measure.

1. BACKGROUND

The lumbar spinal canal consists of 5 interconnected lumbar vertebrae through which the final part of the spinal cord passes and the lumbar and sacral spinal nerves that form the cauda equina. The lumbar canal stenosis can directly affect neurological symptoms and pain in the lumbar region and lower extremities. Due to the frequency of such symptoms, lumbar stenosis has been the subject of research around the world. Many anatomical and radiological morphometric studies have been performed among different populations that show differences in the width of the lumbar canal and the lumbar vertebrae body size within different populations (1-6). Spinal stenosis, a significant orthopedic and neurosurgical disease of the vertebral canal, is a congenital or acquired pathological reduction in the dimensions of the osteo-ligamentous spinal canal that leads to compression of the dural sac and/or spinal nerve roots. The normal mid-sagittal (MSG) diameter of the lumbar spinal canal is greater than 13 mm. If the MSG is from

10 to 13 mm, it is a "relative stenosis", and if the same diameter is less than 10 mm, it is an "absolute stenosis". The normal interpedicular (IP) diameter is between 16 and 18 mm, and a stenotic one is when it is less than 15 mm (7). The use of new radiological imaging methods (CT, MRI), among other advantages, enabled the visualization and work on 3D bone models in a virtual space, as well as precise measurements of individual anthropometric characteristics (8).

2. OBJECTIVE

The objective of this study was to measure, analyze and compare the mediosagital and interpeduncular diameters of the lumbar spinal canal in the population of Bosnia and Herzegovina to other populations around the world.

3. METHODS AND PARTICIPANTS

The study included 200 patients with the MSCT of the lumbosacral spine. The study was carried out at the Radiology Department of the Public Institution General Hospital "Prim. Dr. Abdulah Nakaš" in Sarajevo. It is a retrospective, prospective, comparative and clinical study, where we analyzed and interpreted CT scans of patients with pain in the lumbar spine, who were referred by specialists in various fields. The study included patients whose MSCT scans were performed in the period from January 1, 2013 to December 31, 2014. The subjects with congenital and acquired spinal deformities, patients with benign and malignant changes on the spine and patients with traumatic, inflammatory and pronounced degenerative changes on the spine were not included in the study. A total of 200 patients, 100 female and 100 males, were included in the study. In the group of men, the average age was 43.8 years, with a range from 20 to 60 and a standard deviation of SD=10.99. In the group of women, the average age was 43.78 years, with a range from 27 to 50 years and a standard deviation of SD=6. The CT scan was performed on a 40-slice CT scanner (Siemens Somatom Definition AS) with the following scanning parameters: X-ray tube voltage of 120 kV, current of 266 mAs, Acq 40x0.6 mm and layers 2 mm thick. Axial scans were performed with coronal and sagittal multiplanar reconstructions (MPR), from the 12th thoracic vertebra (TH 12) to the 1st sacral vertebra (S1), bone window (2300-2500 HU) and soft tissue window (300-500 HU). During the analysis of the images, multiplanar reconstructions (MPR) were used, which enabled the examination of one image in the axial, coronal and sagittal planes. Using a bone window, we measured the midsagittal and interpedicular diameters of the spinal canal. Using Siemens Syngo computer software, we measured the following diameters in the axial plane: the midsagittal diameter of the spinal canal was measured in the midsagittal plane, which includes the distance between the back edge of the vertebral body and the front edge of the lamina of the vertebral arch, at the point of separation of the spinous process. The interpedicular diameter of the spinal canal was measured at its widest point, which represents the transverse distance between the inner edges of both pedicles. At the end of the research, we performed statistical processing of the data. We used the SPSS for Windows software package (version 19.0, SPSS Inc, Chicago, Illinois, USA) and Microsoft Excel (version 11. Microsoft Corporation, Redmond, WA, USA) for the statistical analysis of the obtained data.

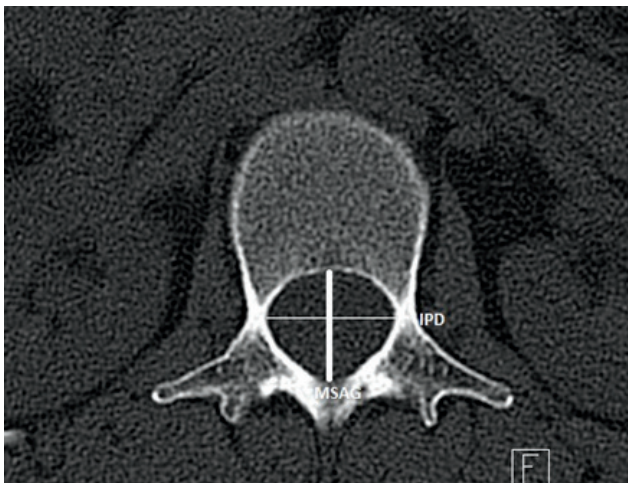


Figure 1. Illustration of the measurement of the lumbar vertebrae diameter. MSGD-midsagittal spinal canal diameter, IPD- interpedicular spinal canal diameter.

4. RESULTS

The study included a total of 200 patients, 100 female and 100 males. In the group of men, the average age was 43.8, with a range from 20 to 60 and a standard deviation of SD=10.99. In the group of women, the average age was 43.78, with a range from 27 to 50 years and a standard deviation of SD=6.

In line with the diagnoses made during the CT scan, it was determined that the leading diagnosis, for both sexes, is disc herniation with 49%, represented by 46% in men and 52% in women. The second most frequently diagnosed diagnosis is

SEX		N	Minimum	Maximum	Mean	Std. Deviation
Male	Age	100	20,00	60,00	43,8400	10,99102
Female	Age	100	27,00	50,00	43,78	6,00

Table 1. Overview of patients by sex

SEX		N	%
Male	Bulged disc	33	33,0
	Disc herniation	46	46,0
	Protrusion	8	8,0
	Extrusion	13	13,0
	Total	100	100,0
Female	Bulged disc	22	22,0
	Disc herniation	52	52,0
	Protrusion	16	16,0
	Extrusion	10	10,0
	Total	100	100,0

Table 2. Overview of patients by diagnosis for both sexes

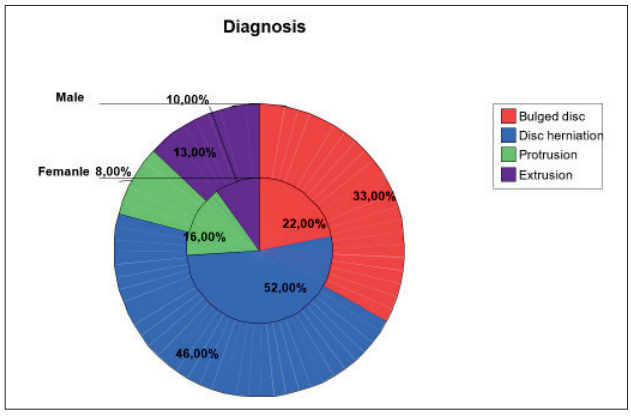


Chart 1. Overview of diagnosis by sex

Vertebra	Male	Female	Both sexes
L1	18,81(15,50-24,30)	19,31(15,50-23,90)	19,06(15,50-24,30)
L2	17,55 (13,00-23,70)	18,55(14,30-22,90)	18,05(13,00-23,70)
L3	16,11(12,30-20,20)	17,20(12,20-21,80)	16,66(12,30-21,80)
L4	16,45(11,70-22,40)	17,38(12,30-23,90)	16,79(11,70-23,90)
L5	17,78(11,90-30,80)	17,83(12,90-26,10)	17,81(11,90-30,80)

Table 3. Overview of midsagittal mean value for both sexes at the level L1 to L5

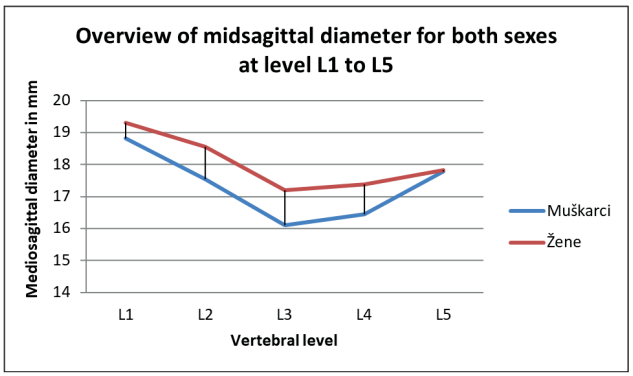


Chart 2. Overview of midsagittal diameter for both sexes at level L1 to L5

bulged disc with a total of 27.5%, and sex-segregated, men account for 33% and women 22%. Protrusion is represented by a total of 12%, whereby in men it is 8%, while in women it is twice as much with 16% share in the total diagnoses of the subjects. Extrusion as the most severe form of degenerative changes on the disc is represented by 11.5% in total, where women are represented by 10% and men by 8%. The determined diagnoses did not make it difficult to precisely analyze the osteo structures and their contours, because they were analyzed with the bone window.

The midsagittal diameters were measured and statistically analyzed. Data are presented as the mean and range for each level for all patients, sex-segregated, and as the total mean. The narrowest diameter was recorded at the L3 level and was 16.66 mm, in both sexes. The range of measurements at all levels was from 11.70 to 30.80 mm. The widest diameter in men was at the level of L1 and was 18.81 mm, while in women it was larger and was 19.31 mm.

Interpedicular diameters were measured from the L1 to L5 level for all patients and data are presented as mean value and range (SD). For the L1 level, the mean value of the interpedicular diameter was measured at 24.78±2.29 mm (range 20.10-33.50 mm), while it was 25.14±2.24 mm (range 20.50-33.20 mm) for L2 level. At the L3 level, the mean value of the diameters was 25.86 ±2.37 mm, (range 21.10-34.60 mm), while it was 27.32 ± 3.06 mm (range 21.60-37.20 mm) at the level of the L4 vertebra. Finally, at the level of the L5 vertebra, the measured diameter is 31.94±4.00 mm (range, 22.10-42.80 mm). The range of interpedicular diameters at all levels was from 20.10 to 42.80 mm. The measurement of the lumbar spine showed a constant increase in the interpedicular diameter from L1 to L5.

Vertebra	Male	Female	Both sexes
L1	25,00(21,30-33,50)	24,56(20,10-31,00)	24,78(20,10-33,50)
L2	25,31(21,70-33,20)	24,96(20,50-31,50)	25,14(20,50-33,20)
L3	26,22(22,20-34,60)	25,51(21,10-32,00)	25,86(21,10-34,60)
L4	27,81(21,60-37,20)	26,84(21,60-34,40)	27,32(21,60-37,20)
L5	32,69(23,50-41,80)	31,20(22,10-42,80)	31,94(22,10-42,80)

Table 4. Overview of interpedicular diameter mean value for male and female, and both sexes L1 to L5 vertebra in mm

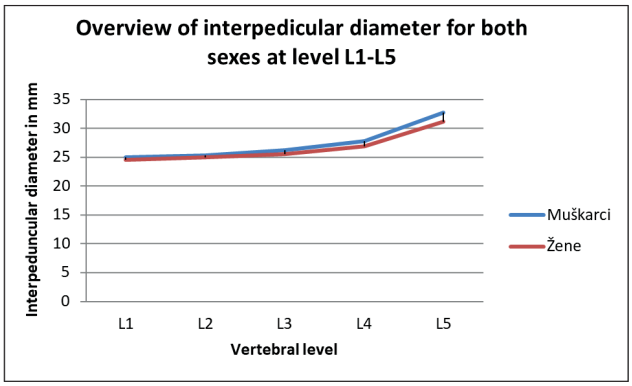


Chart 3. Overview of interpedicular diameter for both sexes at level L1 to L5.

5. DISCUSSION

Midsagittal (MSG) diameter

The widest midsagittal diameter of the lumbar spine spinal canal in both sexes was at the L1 vertebra level and was 19.06 mm (15.50-24.30 mm), while the narrowest was at the L3 level 16.66 mm (12.30-21, 80mm). Proceeding from level L1 to L5, the above-mentioned diameter gradually narrows at levels L3 and L4, only to increase again when descending towards L5 level. At all levels (L1-L5), the midsagittal diameter for the Bosnian-Herzegovinian population is significantly higher in females than in males. Eisenstein S. (15) in his research on the South African population, reported that women

SEX		N	Minimum	Maximum	Mean	Std. Deviation
Male	L1 MSGD canal	100	15,50	24,30	18,8180	1,66905
	L1 IPD canal	100	21,30	33,50	25,0070	2,25242
	L2 MSGD canal	100	13,00	23,70	17,5570	1,97465
	L2 IPD canal	100	21,70	33,20	25,3150	2,21051
	L3 MSGD canal	100	12,30	20,20	16,1180	1,83624
	L3 IPD canal	100	22,20	34,60	26,2200	2,36152
	L4 MSGD canal	100	11,70	22,40	16,4540	2,22354
	L4 IPD canal	100	21,60	37,20	27,8130	3,06468
	L5 MSGD canal	100	11,90	30,80	17,7890	2,84306
	L5 IPD canal	100	23,50	41,80	32,6930	4,18367
Female	L1 MSGD canal	100	15,50	23,90	19,3110	1,57973
	L1 IPD canal	100	20,10	31,00	24,5640	2,33288
	L2 MSGD canal	100	14,30	22,90	18,5520	1,86182
	L2 IPD canal	100	20,50	31,50	24,9690	2,28096
	L3 MSGD canal	100	13,50	21,80	17,2080	2,02113
	L3 IPD canal	100	21,10	32,00	25,5120	2,34242
	L4 MSGD canal	100	12,30	23,90	17,1420	2,15820
	L4 IPD canal	100	21,60	34,40	26,8430	2,99731
	L5 MSGD canal	100	12,90	26,10	17,8310	2,55100
	L5 IPD canal	100	22,10	42,80	31,2020	3,69184

Table 5. Overview of the minimum, maximum and mean values of the midsagittal and interpedicular canals from levels L1 to L5 for both sexes

have a wider midsagittal diameter than men. Similar results were obtained by Malas et al. (17) who came to the conclusion that Turkish women also have a wider MSG diameter than men. In contrast, the Pakistani authors, Alam M et al. (9) in their study reported that the midsagittal diameters in men are larger than in women, but the result is not statistically significant.

By determining the midsagittal diameter of the lumbar spine, from L1 to L5, it was observed in our study that the spinal canal has the shape of an "hourglass" with a narrowing from the level of L1 to L3 and widening from the level of L3 to L5. Lee et al. (16), Eisenstein S. (15) and Amonoo-Kuofi (19) stated that the largest midsagittal diameter is at the L1 level, it gradually decreases, and is the smallest at the L3 level, and then it increases again until the L5 level. Studies by Yadav, Uttam, et al. (9) and Alam, Muhammad M, et al. (11) show a different pattern where there is a progressive narrowing of the midsagittal diameter from the L1 to L4 level and widening at the L5 level. Rema and Rajagopalan (20) observed a gradual decrease in the midsagittal diameter of the spinal canal from L1 to L5. Wang and Shih (21) also reported in their study a progressive decrease in the midsagittal diameter of the spinal canal descending from L1 to L5.

Our study, as well as the vast majority of other previous studies conducted on other populations, confirmed that the widest midsagittal diameter is at the level of the L1 vertebra, while the only exception is a study on the Chinese population (10) that showed the widest midsagittal diameter at the L5 level. The presence of the widest diameter at the L1 level can be explained by the fact that the upper end of the lumbar extension of the spinal cord is exactly at that level, which is the transitional part of the thoracic to the lumbar spine. In humans, this is also the level where thoracic convexity modifies to lumbar concavity, that is, it is the area between the relatively "immobile" thoracic and "mobile" lumbar spine (22). In line with the above, the spinal canal diameter at this transitional level cannot be a reflection only of the size of its content, but also a consequence of the adaptation of this transitional part of the spine with a primarily protective role (19).

The study by American authors Tobin et al. (12) was performed on the American population of Chicago residents (n=918) divided into two age groups, patients born between 1940-1949 (n=546) and patients born between 1970-1979 (n=372). We compared the results of our study with the group born in the period from 1970-1979, since patients from another age group (1940-1949) are older than the population covered by our research. According to the data of the American study, there is a significantly smaller difference in the size of the MSG diameter between men and women at all levels from L1 to L5 compared to the Bosnian-Herzegovinian population from our study. The largest MSG diameter according to the study by Tobin et al. was observed at the L1 level, which positively correlates with the data obtained in our study. The "hourglass" shape starting from level L1 to , with the greatest narrowing at level L3 was also confirmed in the American population included in their study.

Authors Griffith, James F., et al. (10) found in their study of 1080 subjects using the computed tomography that the largest midsagittal diameter is at the level of L5, which is not the case in our study, where the widest midsagittal canal is at the level of L1. A gradual decrease of the same diameter from level L1 to L3 was also described, and then the increase of the aforementioned diameter up to level L5, which is also present in our work.

Vertebra	Our study (male) n=100	Our study (female) n=100	Tobin et al. (male) n=151	Tobin et al. (female) n=221	P-value
L1	18,81	19,31	16,78	16,76	<0,05
L2	17,55	18,55	15,37	15,59	<0,05
L3	16,11	17,20	14,57	14,74	<0,05
L4	16,45	17,38	15,19	15,30	<0,05
L5	17,78	17,83	16,25	16,01	<0,05

Table 6. Comparison of the value of the midsagittal diameter of the vertebra with the study by Tobin et al.

Vertebra	Our study (male) n=100	Our study (female) n=100	J. F. Griffith et al. (male) n=540	J. F. Griffith et al. (female) n=540	P-value
L1	18,81	19,31	15,46	15,58	<0,05
L2	17,55	18,55	14,58	14,87	<0,05
L3	16,11	17,20	14,04	14,18	<0,05
L4	16,45	17,38	15,34	15,03	<0,05
L5	17,78	17,83	19,67	18,12	<0,05

Table 7. Comparison of the value of the midsagittal diameter of the vertebra with the study by J.F. Griffith et al.

Vertebra	Our study (male) n=100	Our study (female) n=100	Uttam Yadu et al. (male) n=174	Uttam Yadu et al. (female) n=128	P-value
L1	18,81	19,31	16,19	16,33	<0,05
L2	17,55	18,55	14,57	14,74	<0,05
L3	16,11	17,20	13,45	13,60	<0,05
L4	16,45	17,38	13,29	13,40	<0,05
L5	17,78	17,83	14,73	14,67	<0,05

Table 8. Comparison of the value of the midsagittal diameter of the vertebra with the study by Uttam Yuda et al.

When comparing the midsagittal diameter values of the current study and diameters in the study by Griffith, James F., et al. there is a statistically significant difference in diameters between the Bosnian-Herzegovinian and Egyptian populations at the L1-L4 levels, but not at the L5 level ($p<0.05$).

Yadav, Uttam, et al. (9) published the study conducted on 302 patients who underwent CT of the abdominal and pelvic organs and who did not experience lumbar pain. The difference in midsagittal diameter width is evident between the study of Yadav, Uttam, et al. (9) and our research at all levels. The measured parameter is significantly higher in the Bosnian-Herzegovinian population than in the analyzed Indian population.

In our study, the midsagittal diameter of the lumbar spinal canal from L1 to L5 has an "hourglass" shape, with the greatest narrowing at the L3 level. Such a pattern was not found in the Indian population in the study of Yadav, Uttam, et al. (9) where the greatest narrowing is at the level of L4, nevertheless the midsagittal diameter of the lumbar spinal canal from L1 to L5 has an "hourglass" shape.

Interpedicular (IP) diameter

In this study, the interpedicular (IP) diameter ranged from 20.10 to 42.80 mm for all lumbar vertebrae from L1 to L5. The smallest IP diameter was at the L1 level and was 24.78mm (20.10mm-33.50mm), while the largest was at the L5 level and was 31.94mm (22.10mm-42.80mm). At all levels from L1 to L5, the IP diameter in men was greater than in women, while it was significantly greater in men at levels L3, L4 and L5.

The interpedicular diameter showed a constant increase from L1 to L5 vertebra. These data are correlated with the data of authors Lee et al. (16) and Zhou et al. (23), who also observed an increase in the IP diameter. Normal IP diameters at all levels were higher for the Bosnian-Herzegovinian population compared to the published data of Egyptian (14), Korean (16), Pakistani (11) and Chinese (10) authors.

In their clinical study, Pakistani authors Alam M et al. examined the lumbar vertebrae using computerized tomography in 49 subjects, and came to the result that the interpedicular diameters are significantly larger at the level of L3 and L5 ($p>0.05$) in men than in women (18), while similar results are also shown in the Pakistani population (11). In 2003, the Turkish authors Tacar et al. (24) published in a morphological study of the lumbar spinal canal, conducted on a sample of 200 patients belonging to healthy adult Turkish population, that in general, at all levels, the mean value of the interpedicular diameter of the spinal canal was greater in men than in women. The difference was statistically significant ($p<0.001$). From L1 to L5, the difference in diameter was 0.5 to 1.5 mm larger in men than in women.

Vertebra	Our study (male) n=100	Our study (female) n=100	J. F. Griffith et al. (male) n=540	J. F. Griffith et al. (female) n=540	P-value
L1	25,00	24,56	20,87	19,83	<0,05
L2	25,31	24,96	21,19	20,13	<0,05
L3	26,22	25,51	22,13	20,96	<0,05
L4	27,81	26,84	24,55	22,95	<0,05
L5	32,69	31,20	29,55	27,41	<0,05

Table 9. Comparison of the value of the interpedicular diameter of the vertebra with the study by J.F. Griffith et al.

Vertebra	Our study (male) n=100	Our study (female) n=100	M.M. Alam et al. (male) n=33	M.M. Alam et al. (female) n=16	P-value
L1	25,00	24,56	24,20	23,50	<0,05
L2	25,31	24,96	24,34	23,46	<0,05
L3	26,22	25,51	24,13	22,36	<0,05
L4	27,81	26,84	24,48	23,81	<0,05
L5	32,69	31,20	28,43	25,96	<0,05

Table 10. Comparison of the value of the interpedicular diameter of the vertebra with the study by M.M. Alam et al.

Vertebra	Our study	Urrutia V.E. et al.	P-value
L1	24,78	22,46	<0,05
L2	25,14	22,80	<0,05
L3	25,86	23,98	<0,05
L4	27,32	26,38	<0,05
L5	31,94	29,78	<0,05

Table 11. Comparison of the mean value for interpedicular diameter of the vertebra with the study by Urrutia V.E. et al.

Mexican authors Urrutia V. et al. (13) examined the morphometric characteristics of the pedicles and bodies of the lumbar vertebrae and published in their paper that the shape of the spinal canal has the appearance of a diamond with a larger transverse diameter increasing progressively from L1 (22.46mm ±1.35) to L5 (29.78mm± 3.50) (21).

When analyzing the available literature, we found inconsistent results between different studies in the world.

In our study, when analyzing all lumbar vertebrae (L1-L5), the interpedicular diameter ranged from 20.10 to 42.80 mm. The smallest diameter was measured at the level of the L1 vertebra, while the largest was measured at the level of the L5. At all levels, the width of the IP diameter was greater in men than in women. The interpedicular diameter showed a constant increase from L1 to L5. The obtained data coincide with the published study of Griffith, J.F. et al. (9), who analyzed interpedicular diameters in the Chinese population. In comparison to the aforementioned study, the IP diameter for the Bosnian-Herzegovinian population is greater at all levels from L1 to L5 in both sexes.

A study on the Pakistani population by M.M. Alam et al. (11) also showed that men of that population have a larger IP diameter at all levels from L1 to L5. In contrast to the data obtained in our study, where the IP diameter progressively increases in both sexes from the level L1 to L5, in the study by Pakistani authors, there is a narrowing of the IP diameter at the L3 level, followed by a widening at the L4 and L5 levels.

Urrutia V.E et al. (13) published in their morphometric study of the pedicles and trunks of the lumbar vertebrae in 60 Mexican subjects, performed with CT, that the smallest IP diameter was recorded at the L1 level and was 22.46 mm, while the largest was at the L5 level 29.78 mm. The measured values of our study for the Bosnian-Herzegovinian population are significantly higher at all levels compared to the Mexican population, where the smallest IP diameter measured at the L1 level is 24.78 mm, while the largest at the L5 level is 31.94 mm.

Below is the comparison of the mean value data for the IP parameter of both sexes in relation to the data provided by the Mexican study.

6. CONCLUSION

Using computed tomography, we measured and calculated the midsagittal (MSG) and interpedicular (IP) diameters of the lumbar vertebrae at the level from L1 to L5 for the population of Bosnia and Herzegovina on a relatively large sample of subjects (n=200). Following the analysis of the results, we concluded that MSG diameter, at all levels of the lumbar spinal canal in the population of Bosnia and Herzegovina is significantly larger in women than in men. The IP diameter of the lumbar spinal canal in the population of Bosnia and Herzegovina at all levels is significantly higher in men than in women. By comparing the obtained results with similar available studies of different world populations, we realized that the dimensions of MSG and IP diameter in both sexes, at all levels of the lumbar spine, in the Bosnian-Herzegovinian population are significantly higher. This information is important since it is necessary to combine clinical and imaging findings to diagnose lumbar spinal stenosis. According to the radiological morphological criteria for CT imaging, a normal

MSG diameter of the lumbar spinal canal is the one whose diameter exceeds 13 mm. If the MSG diameter is between 10 and 13 mm it is a relative stenosis, if the same diameter is less than 10 mm, it is an absolute stenosis. When it comes to the IP diameter of the spinal lumbar canal, values of 16 to 18 mm are taken as the normal diameter value. The stenosis of the spinal lumbar canal is when the IP diameter is less than 15 mm (25). In line with the results of our research, the mean MSG diameter was the narrowest for men at the L3 level and was 16.11 mm, while for women at the same level it was 17.2 mm. The mean value of the smallest MSG diameter for both sexes was 16.66 mm at the L3 level. Therefore, we can conclude that the MSG diameter for both sexes is within the limits of normal values, i.e. greater than 13mm. As for the IP diameter for the population of Bosnia and Herzegovina, the lowest mean value for men was at the level of the L1 vertebra and was 25.0 mm, while for women it was 24.56 at the same level. The mean value of the IP diameter for both sexes was 24.78 mm at the L1 level. Considering the radiological criteria of spinal stenosis of the lumbar spine, where the normal IP diameter is considered to be a diameter with the width of 16 to 18 mm, and the average measured IP diameter of our subjects was 24.78 mm, we can conclude that for our population it is much wider than the set criteria and there is no tendency towards spinal stenosis of the lumbar spine.

- **Acknowledgments:** The authors acknowledge Dr. Tunny Sebastian for her contribution to the statistical analysis.
- **Author's contribution:** All authors were involved in all steps of preparation this article. Final proofreading was made by the first author.
- **Conflict of interest:** This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The authors have no conflict of interest to report.
- **Financial support and sponsorship:** Nil.

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